THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Start with $n = 26$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 10 **(b)** 11 **(c)** 12 **(d)** 9

2. Using Euclid's subtraction algorithm for gcd$(1344, 872)$, the number of subtraction steps needed is: 1

(a) 16 **(b)** 15 **(c)** 13 **(d)** 14

3. Using Euclid's subtraction algorithm for gcd$(912, 720)$, the number of subtraction steps needed is: 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 10

4. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 5702, the final r is: 1

(a) 2075 **(b)** 2084 **(c)** 2066 **(d)** 2165

5. Using gcd$(82, 298) = 2$, the LCM of 82 and 298 is: 1

(a) 12300 **(b)** 12220 **(c)** 12218 **(d)** 12216

6. To test whether 2381 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{2381}\rfloor$. The last number checked is: 1

(a) 49 **(b)** 47 **(c)** 1190 **(d)** 48

7. A counter starts at 0. For i = 1, 2, …, 46, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 23 **(b)** 22 **(c)** 20 **(d)** 21

8. By Euclid's algorithm, gcd$(2624, 1824)$ is: 1

(a) 34 **(b)** 33 **(c)** 31 **(d)** 32

9. **Assertion (A):** The greatest common divisor of 48 and 18 is 6. 1
Reason (R): The gcd of two numbers is the largest number that divides both of them.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** $29 \bmod 7 = 1$. 1
Reason (R): $7 \bmod 29 = 7$.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the gcd of 380, 520 and 152 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 2

12. A floor of 402 cm by 318 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's subtraction algorithm to find the tile side and the number of tiles. 2

13. Find the gcd of 344 and 100 by Euclid's subtraction algorithm and use it to find their LCM. 2

14. Find the gcd of 280, 480 and 150 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Trace: count = 0; for i = 1 to 11: for j = i to 11: count = count + 1. Find the final count and explain the pattern. 3

16. Trace: total = 0; for i = 1 to 11: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

17. Start with n = 23. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

OR

Trace: count = 0; for i = 1 to 6: for j = i to 6: count = count + 1. Find the final count and explain the pattern. 3

18. Trace: total = 0; for i = 1 to 8: if i is a multiple of 4, add i^2 to total. List the numbers added and the final total. 3

OR

Start with n = 20. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ananya adds $586 + 947$ column by column, starting from the ones column, and records the carry from each column. 5

(i) What digit is written and what is carried from the ones column?
(ii) What digit is written and what is carried from the tens column?
(iii) What digit is written in the hundreds column, and what is the carry?
(iv) Write the final sum and the total number of carries produced.

20. Find the largest number that divides both 340 and 250 exactly, using Euclid's subtraction algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 864, 320 and 212 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 5

END OF QUESTION PAPER — Total: 40 Marks